

Molecular Simulation

The aim of this project is to estimate the transport properties of the propylene oxide-water system in a tubular plug flow reactor (PFR). The priority property of this model is selected as the diffusivity of propylene oxide (PO), which is a key factor that defines the transport properties and thermodynamics of molecules in the PFR. Molecular Dynamics (MD) was chosen as the main simulation method, as it is more suited to evaluate time-dependent transport properties (diffusivity), than Monte Carlo, which is known to be more suited for sampling equilibrium properties.

Method

The model was developed with an increase in complexity, from pure water and pure propylene oxide to the binary mixture. A Coarse-Grained (CG) molecular model and the **Mie Potential** were utilised to simplify the complex shapes of PO and water by grouping atoms into “super-atoms” beads (Herdes, Totton and Müller, 2015). The Mie potential parameters (λ , ϵ and σ) were derived from SAFT using critical conditions and the acentric factor of the compound as shown in the **Appendix** (Table 3). The λ , ϵ and σ used for water and propylene oxide are shown in Table 2, the parameters for water were obtained from the literature data to ensure accuracy (more explanation in the **Appendix**).

Table 1. Simulation results vs literature results

	Simulation Result	Literature Value	Source for literature
Diffusivity	$7.0196 \times 10^{-9} \text{ m}^2/\text{s}$	$1.1961 \times 10^{-9} \text{ m}^2/\text{s}$ (PO in H ₂ O) $2.2952 \times 10^{-9} \text{ m}^2/\text{s}$ (H ₂ O in H ₂ O)	(Yaws, 2010), (Eastal, Price and Woolf, 1989)
Density	1021 kg/m^3	830 kg/m^3 (PO), 1000 kg/m^3 (H ₂ O)	CFD brief
Isochoric specific heat capacity	2.1076 J/(molK) (MD) 17.39 J/(molK) (MC)	54.8 J/(molK) (H ₂ O)	(The Engineering ToolBox, 2004)

The molecules were modeled as 216 beads in a cubic box with Periodic Boundary Conditions (PBC) to mimic an infinite bulk of the fluid mixture. Particle interactions were calculated at the closest image using the Minimum Image Convention (MIC), while the trajectories were computed using the velocity Verlet integration loop. A binary mixture was modeled with a composition of 95% water and 5% PO. The Lorentz-Berthelot mixing rules were implemented to estimate cross-interaction parameters (λ_{ij} , ϵ_{ij} , σ_{ij}), which were stored in matrix form to define all pairs of interactions (water-water, PO-PO, and PO-water). GROMACS units (kJ/mol, nm, ps) were utilised to ensure consistency of unit conversions. The box length ($L=1.9 \text{ nm}$) derived from the number of molecules ($N=216$), was larger than of larger than twice the cut off radius ($2r_c=1.5 \text{ nm}$). The system was first equilibrated at 298.15 K and 1 bar in the NPT ensemble using the Berendsen thermostat and barostat to determine the mixture density, while allowing the box volume to fluctuate. Finally, a production run was performed in the NVT ensemble, the Mean Square Displacement (MSD) was calculated on-the-fly from unwrapped trajectories, and the diffusivity was extracted via the Einstein relation.

Results & Discussion

As shown in Table 1, the PO diffusivity $7.0196 \times 10^{-9} \text{ m}^2/\text{s}$ was obtained from the main simulation, which falls within the acceptable range for liquid transport of 10^{-10} to $10^{-9} \text{ m}^2/\text{s}$ (COMSOL, 2017). This diffusivity value is at least a factor of 6 higher than the literature value ($1.196 \times 10^{-9} \text{ m}^2/\text{s}$), due to the loss of microscopic friction during the use of CG, and hence, will not be used further in the CFD model. The predictive capacity of the model was evaluated by varying the temperature from the baseline 298K. The model successfully predicted the fluid's Arrhenius behaviour as the diffusivity increased to $8.390 \times 10^{-9} \text{ m}^2/\text{s}$ at 350K and decreased to $4.656 \times 10^{-9} \text{ m}^2/\text{s}$ at 240K. The mixture density was found to be higher than the density of both pure PO and H₂O, exhibiting the non-ideal solution behaviour, where uneven interactions led to the reduction in volume. A Monte Carlo model was implemented for a learning purpose, using the same potential and parameters, it generated a similar total energy compared to the MD ($\sim 4000 \text{ kJ/mol}$). However, the heat capacity estimated from energy fluctuation in both MD and MC are found to be much lower than the literature value due to the limitation of the CG and the Berendsen thermostat, which filtered out detail from fluctuations. Lastly, radial distribution shows that there are no molecules interactions within the distance closer to $r = 0.3 \text{ nm}$, representing the repulsive nature of the Mie potential. In addition, the timestep was set to 0.001 ps to prevent the energy drift in the MSD curve.